

Quiz — 1.3.1 Compression, Encryption and Hashing

Name: _____ Date: _____ Mark: _____ / 25

OCR H446 · Component 01 · 1.3.1

Section A — Multiple choice (8 marks)

1. Which statement about lossy compression is correct? A. The original file can always be perfectly reconstructed B. Data is permanently removed and cannot be recovered C. It is only used for text files D. It never reduces file size

Your answer: ☐

2. Run Length Encoding works best on data that contains: A. Random values with no repetition B. Long runs of repeated identical values C. Encrypted ciphertext D. A single unique value per pixel

Your answer: ☐

3. A disadvantage of RLE is that it can: A. Make files larger when there are few repeated runs B. Permanently lose data C. Only compress audio D. Require a public and private key

Your answer: ☐

4. In dictionary-based compression, what must be supplied so the file can be decompressed? A. The encryption key B. A hash digest C. The dictionary (index of patterns) D. The damping factor

Your answer: ☐

5. In symmetric encryption: A. A public key encrypts and a private key decrypts B. The same key is used to encrypt and decrypt C. No key is needed D. The key is never shared but is also never used

Your answer: ☐

6. Which is the main weakness of symmetric encryption that asymmetric encryption solves? A. It is too slow to be practical B. The secure distribution of the shared key C. It produces a fixed-length output D. It cannot encrypt text

Your answer: ☐

7. A digital signature is created by: A. Hashing the message with no key B. Signing with the sender's private key (verified with the public key) C. Encrypting with the recipient's public key only D. Using RLE on the message

Your answer: ☐

8. Which property means a hash function always produces the same digest for the same input? A. One-way B. Collision-resistant C. Deterministic D. Reversible

Your answer: ☐

Section B — Short answer (12 marks)

9. Define lossless compression and give one suitable use. [2]

10. Using RLE, encode the following pixel run. Write your encoded output as value+count pairs. [2]

W W W W W B B W W W W W W

11. The following data was encoded with RLE as 5A 3B 1C 4A . Decode it back to the original sequence. [2]

12. Explain why asymmetric encryption is described as more secure for exchanging data with a stranger than symmetric encryption. [2]

13. A website stores user passwords by hashing them. Explain two reasons why hashing (rather than encryption) is used. **[2]**

14. State two properties a good hash function should have. **[2]**

Section C — Exam-style question (5 marks)

15. A music streaming service sends audio files to users' devices and stores users' login details on its server.

Describe how compression, encryption and hashing could each be used by this service. Refer to the type of compression suitable for audio, the type of encryption used during data transfer, and how hashing protects stored login details. **[5]**
